

# AS Level Mathematics — Pure Mathematics 1 (P1)

## Practice Paper 4

Time allowed: **1 hour 50 minutes**

### Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [ ] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

### List of formulae (as provided in the examination)

Quadratic equation: for  $ax^2 + bx + c = 0$ ,  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion:  $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$ , where  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series:  $u_n = a + (n-1)d$ ,  $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series:  $u_n = ar^{n-1}$ ,  $S_n = \frac{a(1-r^n)}{1-r}$  ( $r \neq 1$ ),  $S_\infty = \frac{a}{1-r}$  ( $|r| < 1$ )

Circular measure: arc length  $= r\theta$ , sector area  $= \frac{1}{2}r^2\theta$  ( $\theta$  in radians)

Trigonometry:  $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$ ,  $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration:  $\frac{d}{dx}(x^n) = nx^{n-1}$ ,  $\int x^n dx = \frac{x^{n+1}}{n+1} + c$  ( $n \neq -1$ )

**1** Find the coefficient of  $x^2$  in the expansion of  $\left(2 - \frac{x}{2}\right)^8$ . [3]

**2** Solve the equation  $2 \sin \theta \tan \theta = 3$  for  $0^\circ \leq \theta \leq 360^\circ$ . [4]

**3** A ladder has 20 rungs. The lengths of the rungs decrease uniformly from the bottom rung to the top rung, so that the lengths form an arithmetic progression. The bottom rung is 60 cm long and the top rung is 41 cm long.

**(a)** Find the difference in length between consecutive rungs. [2]

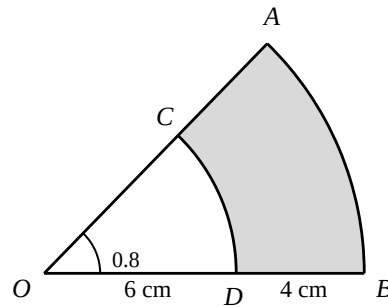
**(b)** Find the total length of the 20 rungs. [2]

4 The points  $A$  and  $B$  have coordinates  $(2, 7)$  and  $(6, -1)$  respectively.

(a) Find the equation of the line  $AB$ , giving your answer in the form  $y = mx + c$ . [2]

(b) The point  $C$  lies on the  $x$ -axis and is equidistant from  $A$  and  $B$ . Find the coordinates of  $C$ . [3]

5



The diagram shows a sector  $OAB$  of a circle with centre  $O$  and radius  $10$  cm, where angle  $AOB = 0.8$  radians. The points  $C$  and  $D$  lie on  $OA$  and  $OB$  respectively, with  $OC = OD = 6$  cm, and  $CD$  is an arc of a circle with centre  $O$ . The region  $ABDC$  is shaded.

(a) Find the perimeter of the shaded region. [3]

(b) Find the area of the shaded region. [3]

6 Functions  $f$  and  $g$  are defined for  $x \in \mathbb{R}$  by

$$f(x) = 3x - 2, \quad g(x) = x^2 + 4x.$$

(a) Find and simplify an expression for  $gf(x)$ . [2]

(b) Solve the equation  $gf(x) = 32$ . [2]

(c) Explain why  $g$  does not have an inverse. The domain of  $g$  is now restricted to  $x \geq k$ . State the least value of  $k$  for which  $g^{-1}$  exists. [2]

7 The line  $y = mx + 2$  and the curve  $y = x^2 - 4x + 6$  are given, where  $m$  is a constant.

(a) Find the two values of  $m$  for which the line is a tangent to the curve. [4]

(b) For each of these values of  $m$ , find the coordinates of the point at which the line touches the curve. [3]

**8** The equation of a curve is  $y = x^3 - 3x^2 - 9x + 5$ .

- (a) Find the set of values of  $x$  for which  $y$  is a decreasing function of  $x$ . [4]
- (b) Find the equation of the normal to the curve at the point where  $x = 0$ , giving your answer in the form  $ax + by + c = 0$ . [3]

**9**

- (a) A curve is such that  $\frac{dy}{dx} = 3\sqrt{x} - \frac{1}{x^2}$  and the point  $(1, 5)$  lies on the curve. Find the equation of the curve. [4]
- (b) Find the area of the region bounded by the curve  $y = x^2 + 1$ , the  $y$ -axis and the line  $y = 5$ . [4]

**10**

- (a) The second term of a geometric progression is 6 and the fourth term is 54. All the terms of the progression are positive. Find the sum of the first 8 terms. [4]
- (b) An arithmetic progression has common difference 2. The 1st, 3rd and 7th terms of this arithmetic progression are the first three terms of a geometric progression. Find the first term of the arithmetic progression and the common ratio of the geometric progression. [4]

**11**

- (a) Prove the identity  $(1 - \cos \theta) \left( 1 + \frac{1}{\cos \theta} \right) \equiv \frac{\sin^2 \theta}{\cos \theta}$ . [3]
- (b) Hence solve the equation  $(1 - \cos \theta) \left( 1 + \frac{1}{\cos \theta} \right) = \frac{3}{2} \sin \theta$  for  $0^\circ < \theta < 360^\circ$ . [4]
- (c) Solve the equation  $2 \sin \left( x + \frac{\pi}{6} \right) = 1$  for  $0 \leq x \leq 2\pi$ , giving your answers in terms of  $\pi$ . [3]

**12** A circle has equation  $x^2 + y^2 - 4x + 6y = 12$ .

- (a) Find the coordinates of the centre and the radius of the circle. [2]
- (b) Find the coordinates of the points where the circle meets the  $x$ -axis. [3]
- (c) The line  $y = 1$  meets the circle at the points  $P$  and  $Q$ . Find the length of the chord  $PQ$ . [2]

---

**END OF PAPER** (Total: 75 marks)